

Towards intelligent extended reality in stroke rehabilitation

Application of machine learning and artificial intelligence in rehabilitation

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15.1 Introduction

Stroke is the leading cause of acquired disability worldwide causing sensory, motor and cognitive impairments (Global, regional, & national burden of stroke, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016, 2019). Poststroke recovery, which depends on neuroplasticity, is variable, often incomplete and up to 55% to 75% of the surviving patients are left with some residual deficit at 3 and 6 months (Duncan et al., 1994). The overarching aim of poststroke rehabilitation is to augment the putative mechanisms of plasticity via targeted therapy to enhance the recovery of function. There is emerging evidence that carefully directed, well focused therapy of longer duration increases the “dose” of rehabilitation and improves outcome (Krakauer & Carmichael, 2017). The dose-dependence of recovery can be defined as:

Restitution/Recovery (Magnitude) = Behavior (dose) X Representation (Residual amount) X Plasticity (Level) (Krakauer & Carmichael, 2017).

Several recent metaanalyses suggest that larger doses (> 2 hours per day) lead to clinically meaningful improvements of the upper limb poststroke (Schneider, Lannin, Ada, &

Schmidt, 2016). However, the availability of therapists is a significant bottleneck in stroke rehabilitation and patients spend over 80% of their daytime in nontherapeutic activities (Bernhardt, Dewey, Thrift, & Donnan, 2004). Despite the National Institute of Health and Social Care (NICE) recommendation in the UK for 45 minutes rehabilitation therapy, five days a week, most UK stroke units are unable to comply and patients receive an average of only 16 minutes of each type of therapy. There is an urgent need for novel rehabilitation strategies to improve poststroke outcomes.

The use of technology that combines real and virtual world interaction is known as Extended Reality (XR). This broad term encompassed Virtual Reality (VR), Augmented Reality (AR), wearables where a user can interact with computer generated environments. XR systems have created a new paradigm that enables healthcare providers to augment the dose of task-oriented exercises delivered both in the clinic and, via telerehabilitation models, in the home. This allows therapy time to be increased without recruiting additional trained therapists. Combining XR with Artificial Intelligence (AI) and/or Machine Learning (ML) will further reduce the level of supervision required, enabling therapy to be provided by less-skilled staff or family members.

15.2 Current methods of stroke rehabilitation

Effective rehabilitation is the key to helping stroke survivors live meaningful, occupationally rich and satisfying lives. The theoretical foundation is based on the World Health Organization's (WHO) international classification of functioning (ICF). WHO defines rehabilitation as:

“A set of interventions designed to optimize functioning and reduce disability in individuals with health conditions in interactions with their environment” (Shakespeare & Officer, 2011).

Neuro-rehabilitation is an integral element of poststroke care today. The individual needs of stroke survivors is such that no single profession can carry out this rehabilitation alone. Client-centered rehabilitation practice is delivered from a harmonized and integrated group of health and social care professions. These include Occupational Therapists, Physiotherapists, Speech and language therapists, Nurses, Physicians, Psychologists, Orthoptists, Social workers, and Dieticians all sitting within an interdisciplinary team.

Evidence shows that the most effective rehabilitation is carried out within a multidisciplinary stroke unit. Langhorne et al.'s (2020) *metaanalysis* confirms with moderate quality evidence that stroke patients admitted to an inpatient stroke unit are more likely to be alive, independent, and living at home one year after stroke (Langhorne & Ramachandra, 2020).

Rehabilitation starts early after the initial 24 hours following stroke (Langhorne, Wu, Rodgers, Ashburn, & Bernhardt, 2017) but the process can be lifelong with differing levels of professional input at each phase. A person's insight and ability to engage with therapy, life experiences, and changing circumstances mean the entire rehabilitation journey fluctuates in intensity. Activities take place in numerous settings including hospital, stroke units, outpatient clinics, and community or within the home environment. The focus of rehabilitation is always client-centered and preferably occupation-centered. Motivational goals are created by the person and their family with the aim of maximizing well-being and societal participation.

An important component is undoubtedly therapy input. It is useful at this stage to consider the definition of “therapy” provided by WHO:

“Therapy is concerned with restoring and compensating for the loss of functioning, and preventing, or slowing deterioration in functioning in every area of a person’s life” (Shakespeare & Officer, 2011).

In the United Kingdom, the Royal College of Physicians National Clinical Guidelines for Stroke 2016 (NICE accredited) provide the most user-friendly and up-to-date document on how stroke care should be provided (Royal College of Physicians London, 2016) to guide clinical practice with the latest evidence and ensure the highest quality of care. There is evidence in support of some specific treatments recommended within the guidelines. For example, a Cochrane systematic review found that people with stroke who received occupational therapy targeting Personal Activities of Daily Living (PADL), an important component of activities of daily living (ADL), performed better and had a reduced risk of a poor outcome (dependency in PADL, deterioration or death) compared to those without occupational therapy input (Legg, Drummond, & Langhorne, 2006). It needs to be mentioned that these guidelines apply to a high income country with specific organizational structures supporting the recommendations. Work is currently being carried out by international clinicians and academics to implement international guidelines that better represent therapeutic approaches based on the best quality evidence rather than organizational issues (WHO, 2020).

Therapy involves a continuous process of assessment, goal setting, treatment interventions and continued reassessment by qualified therapists. Impairments in motor functioning, sensation, cognition, vision, speech and language and swallowing are commonly screened initially so the individualized therapy can be initiated. Measures such as exercise programs, repetitive task training through purposeful activities of daily living (ADL) practice and compensation strategies including appropriate modifications to the environment are employed to support rehabilitation. Supporting, counseling, and educating stroke survivors and their families are integral elements. Trans-cranial magnetic stimulation, Assistive Technology, Robotics, and Virtual Reality show great potential but are not yet widely available in all stroke units. We will consider some but by no means all of the current therapy measures within the scope of this chapter and have mainly focused on inpatient rehabilitation (Table 15.1; based on NICE 2016 guidelines).

Continence, hydration, nutrition, swallowing, fatigue, driving, conjugal life, mental capacity, mood cognition, and mouth care are all integral within the rehabilitation pathway alongside the contents of Table 15.1. Therapy focuses on ensuring a person can maximize their participation in all activities of daily living and social participation. Practice opportunities, equipment, adaptations and training with a 24-hour approach to therapy are the key elements. No therapy session and intervention combination is the same because the approach is individualized rather than solely impairment driven.

In the UK, it is evident the multidisciplinary team is not yet meeting the needs of all stroke survivors effectively. Despite 45 minutes of each type of therapy per day being advised by the Royal College of Physicians, recent audit data demonstrates that the average amount of therapy ranged from 2 minutes (psychology) to 14 minutes (physiotherapy) per day of

Table 15.1 Current methods of stroke rehabilitation therapy.

Motor functioning	Pain, spasticity, sensation	Vision	Cognition and perception	Mood	Communication	ADL
Assess for motor impairment or <i>ataxia</i> and assess <i>Mobility</i> . Mobilization within 24 hours of onset only for patients who require little or no assistance. Task-specific, repetitive, intensive exercises or activities that will increase muscle strength. Treadmill training or walking interventions at higher intensity than usual care. Ankle foot orthosis. Functional Electrical Stimulation for gait improvement. Assess for walking aids and wheelchair.	Avoid shoulder subluxation. Careful de-weighting and positioning of the arm. Educate the family to handle correctly and avoid mechanical stress and excessive movement. Monitor pain. Offer analgesia.	Assess vision including visual field.	Routine cognitive screening to identify level of functioning. Adjustments to multidisciplinary treatments to enable participation. Interventions aimed at developing compensatory behaviors and adaptive skills. Input from neuropsychologist. Education for family and carers <i>Apraxia</i> screening. Task specific practice of activities in own environment	Assess for mood disorder. Psychological interventions such as motivational interviewing.	Assessment by Speech and Language Therapist Practice communication as frequently as tolerated in the first 4 months post stroke. For dysarthria training in techniques to improve clarity of speech. For speech apraxia training to maximize articulation of key words to improve speech intelligibility	Practice self -care and dressing. Practice kitchen activities. Meal and hot drink preparation. Practice shopping, purchasing, multiple errands and social interaction.
Assess <i>Arm function</i> . Constraint- induced movement therapy. Opportunities to practice functional activities. Education on care and monitoring of affected upper limb. Mental imagery.	Assess for <i>spasticity</i> and risk of contractures. Positioning, active and passive movement. Pain control. Assess a therapeutic goal for Botulinum toxin. Refer to spasticity service. Therapy and splinting as appropriate 12 weeks after injections. Antispasticity drugs	Orthoptist assessment and information/ education	Assess for <i>dis-executive syndrome</i> . Train compensatory techniques using internal and external strategies Attention training exercises and teaching strategies to improve attention skills,	Antidepressant medication	Educate family to support. Computer based practice. Functional treatment. Educate the health team, carers and family to help optimize engagement in rehabilitation, and promote autonomy and social participation.	Training on use of equipment and adaptations for safe independence

<i>Assess sitting and standing balance.</i> Trunk training exercises. Lower limb exercises. Functional task specific training. Balance training.	<i>Sensory impairment</i> screen. Mirror therapy. Electrical stimulation. Sensory program.	Practice compensation strategies in daily living tasks in a safe environment. Environmental adaptations as appropriate.	<i>Memory</i> assessments. Training to help encode, store and retrieve. Train to use compensatory techniques Education for family and carers	Increased exercise, social interaction and educational groups.	Assistive technology and communication aids to supplement or compensate for limited speech.	Education for families and carers in how to provide support
<i>Falls</i> training. Educate and train how to get up after a fall. Assessment of fragility fracture risk. Participate in exercise for balance and coordination.	Review <i>central poststroke pain</i> Assess physical and psychological well-being, Monitor impact on lifestyle, sleep, activities and participation, and the continued need for pharmacological treatment	Education regarding driving restrictions	<i>Perceptual</i> assessment. Education. Make the environment safe. Task repetition practice. Sensory stimulation. <i>Spatial awareness/ visual inattention</i> assessment. Visual scanning training. Prisms. Mirror therapy. Sensory stimulation. Prompts. Compensatory strategies. Education	Assess for <i>emotionalism</i> . Distract from stimulus for emotionalism Antidepressant medication.	Referral to aphasia groups as appropriate	

inpatient stay ([Gittins, Vail, & Bowen, 2020](#)). A major challenge is that staffing levels differ within teams and intensity of therapy provision varies between hospitals and community services making level of therapy provision often dependent on postcode.

Ensuring that early supported discharge is facilitated by the stroke team is another challenge for the rehabilitation service. Patients being discharged early should receive the same intensity of therapy and multidisciplinary interventions available in hospital. It should not result in a delay in delivery of therapy but the reality is that this recommendation is rarely consistently achieved. The efforts to create a seamless service in this transition from hospital to home are made more difficult by varying staffing levels and complexities of individual needs. Ensuring the right level of resources are available for rehabilitation in the home environment is crucial. Arguably the most challenging aspect of providing stroke rehabilitation is getting the intensity and dosage right. Time constraints due to staffing, organizational, social and environmental issues can be barriers to providing sufficient therapy at the right time during stroke recovery.

15.3 Current extended reality technologies in stroke rehabilitation

The application of XR for rehabilitation and therapy has been the focus of recent research activity to determine whether it can be used to augment conventional techniques for greater engagement, increased intensity and improved patient outcomes.

Although research has mainly focused on motor rehabilitation, work has also been done in cognitive rehabilitation and speech therapy. A Cochrane report published by Laver et al., in 2017 was designed to determine the efficacy of VR compared to alternative or no intervention on upper limb function and activity, as well as a variety of other rehabilitation activities ([Laver et al., 2017](#)). This was based on 72 published studies involving 2470 stroke survivors. The report found evidence that the use of VR and interactive gaming was effective when used in conjunction with normal care routines. We summarize below relevant work in the application of VR to stroke rehabilitation from the last three years, and highlight the emergence of AR and other XR-based solutions. During this period affordable head-mounted displays have become widely available, making highly immersive VR applications possible, alongside a first wave of AR-specific headsets.

15.3.1 Motor rehabilitation

In 2017, Dascal et al. conducted a systematic review of randomized controlled trials that examined VR applications across many acute inpatient medical settings ([Dascal, Reid, & Ishak, 2017](#)). This found that VR was a promising intervention with applications in pain management, eating disorders, motor, and cognitive rehabilitation. It included some evidence of efficacy in VR rehabilitation, but larger studies are needed to show clinical and cost-effectiveness. A review by Kim et al., investigated the application of VR to motor rehabilitation after stroke, focusing primarily on upper limb treatment ([Kim, Cho, & Ku, 2020](#)). VR was

shown to actively engage patients in the rehabilitation program leading to better motor-recovery outcomes. While current evidence shows VR is beneficial in conjunction with standard rehabilitation, it is unclear who would benefit most, which type of VR, immersive or nonimmersive, is more effective. Tieri et al., conducted a review looking at the scientific rationale behind VR for future rehabilitation (Tieri, Morone, Paolucci, & Iosa, 2018). They posed the question of why patients should perform rehabilitation in a virtual environment and not a real environment. They concluded that of the advantages and new possibilities that VR offers, the most promising is the ability to develop multitasking rehabilitation scenarios in a virtual environment. Boone et al., conducted a study to evaluate acceptability, recruitment, and retention rate of VR-assisted rehabilitation and to determine which outcome measures best capture the effect of the intervention (Boone, Wolf, & Engsberg, 2019). Results showed that therapy using their MetacogVR system was perceived as acceptable and feasible for adults with chronic stroke. Upper extremity motor performance, occupational performance, and participation were deemed as the best outcome measures to capture the effects of the system.

There is emerging use of AR in motor rehabilitation where virtual objects can be superimposed onto a real world scene. Assis et al., discuss a feasibility study using AR for upper-limb poststroke motor rehabilitation (de Assis, Corrêa, Martins, Pedrozo, & Lopes, 2016). Escalona et al., propose a system called EVA, an At-Home rehabilitation system using AR and low-cost sensors (Escalona, Martinez-Martin, Cruz, Cazorla, & Gomez-Donoso, 2020). The system is designed as an AR platform to help engage and supervise at-home rehabilitation sessions using low-cost sensors, as well as to store statistics that allow therapists to tailor exercises based on past performance. Results presented from the study showed promising outcomes for the AR system.

15.3.2 Speech therapy

There has been less research conducted into the application of XR in speech therapy with Stroke patients. Giachero et al. present a study that used VR to assist with aphasia rehabilitation (Giachero, Calati, & Pia, 2020). They found no significant difference between the conventional therapy group and the conventional therapy with VR support group. However in the VR group, the improvement shown was distributed over more cognitive and psychological aspects compared to the control group. Marshall et al., investigated whether it was feasible to deliver group social support to people with aphasia via a multiuser, virtual platform (Marshall, Devane, & Talbot, 2020). The study consisted of 14 sessions over 6 months and results suggested that a larger trial of remote group support using VR would be merited. However, the treatment content and regime used within the study, as well as the outcome measures used, should be reviewed before conducting another trial.

There has also been research conducted in other areas that could be relevant to stroke survivors, such as speech therapy for children with autism. da Silva et al., presented a study that discussed the integration of AR in communication interventions for children with Autism Spectrum Disorder (da Silva, Fernandes, & Grohmann, 2015). This was achieved by

relating elements in the AR environment with alternative communication and applied behavior analysis strategies. The study found that this kind of approach has added value in speech therapy and, when coupled with traditional methods, is an option that should be explored further.

15.3.3 Cognitive rehabilitation

In 2019, Maggio et al., conducted several different reviews into VR in cognitive rehabilitation. One paper reviewed the use of VR tools in cognitive rehabilitation of stroke-affected individuals (Maggio, Latella, & Maresca, 2019). They concluded that these could positively affect stroke patient cognitive outcomes by boosting their motivation and participation in the therapy sessions. Another evaluated the role of VR in cognitive rehabilitation of different neurological diseases and the accessibility to healthcare systems providing VR treatment (Maggio, Maresca, & De Luca, 2019). The review stated that VR could positively affect neurological patients' outcomes by boosting motivation and participation, enhancing conventional therapies, promoting longer sessions and a reduced overall hospitalization time. Riva et al., reviewed VR in neuro-rehabilitation to analyze its effects on memory, attention, executive functions, language, and visuospatial ability (Riva, Mancuso, Cavedoni, & Stramba-Badiale, 2020). They found that VR-based neuro-rehabilitation showed encouraging results for executive functions and visuospatial abilities, particularly for acute and neurodegenerative conditions.

Cameirão et al. investigated the impact in stroke recovery of a virtual cognitive-motor task customized with positive stimuli in comparison to standard rehabilitation (Cameirão, Pereira, & Badia, 2017). They found that both groups showed significant improvements over time in functional ability during task performance but without changes in motor impairment. There were no significant differences between the groups at the end of treatment and follow-up, and the impact of VR therapy was lower than in similar studies with stroke patients without cognitive deficits. In 2016, Faria et al., conducted research into the benefits of using VR for cognitive rehabilitation (Faria, Andrade, Soares, & i Badia, 2016). They used a VR-based system called Reh@City for their intervention, which incorporated a virtual simulation of a city that allowed the patients to practice activities of daily living that helped improve their memory, attention, visuo-spatial abilities, and executive functions. Faria et al., later investigated the feasibility for stroke recovery of a virtual cognitive-task, Reh@Task (Faria et al., 2018). This updated VR system combined adapted arm reaching, and attention and memory training tasks. Results from their study confirmed the viability of VR tools that combined motor and cognitive training. John et al., present ongoing work on the development of a VR environment for cognitive rehabilitation of stroke patients as part of their recovery (John, Day, & Pop, 2019). The system is designed to allow patients to practice an ADL, assisted by a therapy assistant or family members, using immersive virtual technologies. For example, in Fig. 15.1 the patient is counting out coins to pay for an item. It also tracks the patient's performance and allows the clinicians to adjust the difficulty of the scenarios to help with the patient's cognitive recovery. Early results indicate that patients with

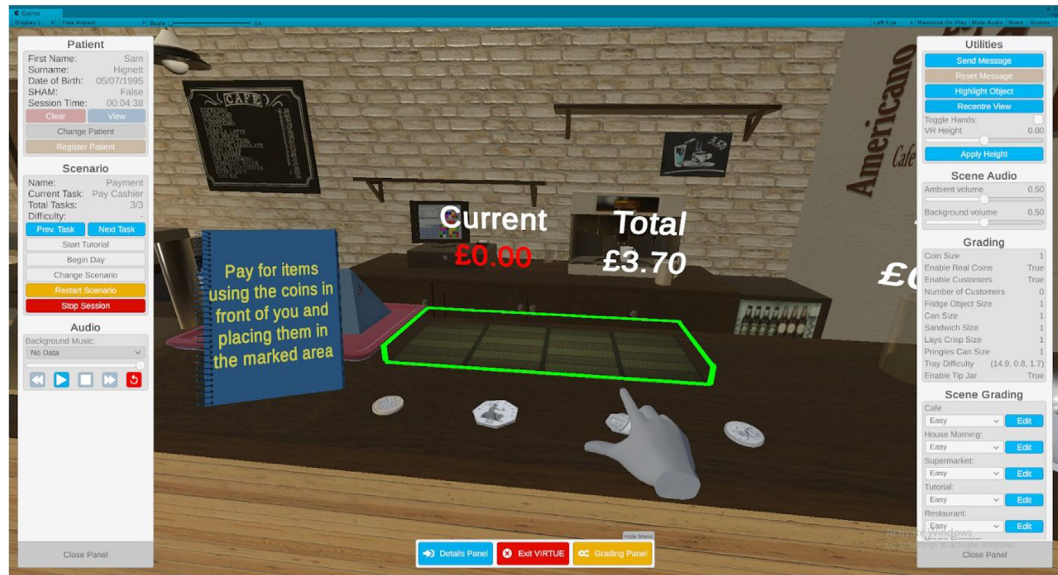


FIGURE 15.1 Virtual reality (VR) application in stroke rehabilitation. The activity of counting coins as an activities of daily living (ADL). An immersive VR application being used for the cognitive rehabilitation of stroke survivors.

poor cognition following a stroke are more likely to show a better improvement when exposed to VR therapy.

Fig. 15.1. A counting coins ADL from an immersive VR application being used for the cognitive rehabilitation of stroke survivors (John et al., 2019).

AR in cognitive rehabilitation is also a field of research that is receiving more attention. Song et al., developed a mobile phone-based AR rehabilitation system for stroke rehabilitation (Song, Ding, Zhao, Jia, & Shull, 2019). They implemented three serious games for upper-limb motor function and cognitive training in which the patient had to move their arms to touch virtual targets generated in the virtual environment. Results showed that motor performance improved over time, and most patients felt an improved mental state while playing the AR rehabilitation games. A crossover study by Rohrbach et al., used an AR approach with a Microsoft HoloLens to support patients in a tea making task (Rohrbach, Gulde, & Armstrong, 2019). Participants received three-dimensional dynamic holograms of the substeps necessary to complete the task. Conclusions showed that the approach was feasible with high patient- acceptance, although advances in hardware and patient-interface are necessary to assist patients suffering from a cognitive impairment.

15.3.4 Machine learning for improving extended reality

Machine learning comprises three subareas—supervised, unsupervised, and reinforcement learning (Fig. 15.2). Supervised learning, also referred to as predictive learning, is an approach

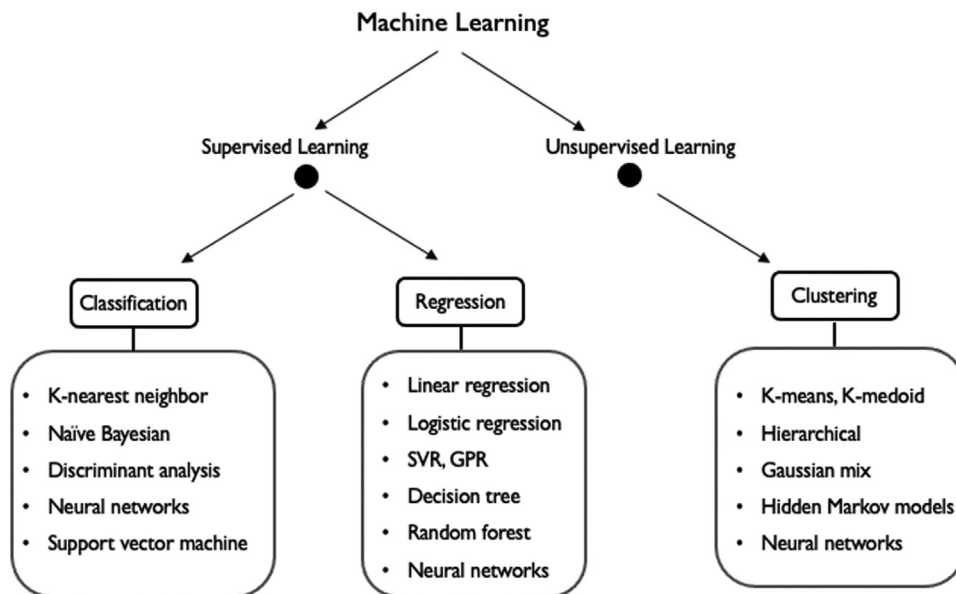


FIGURE 15.2 Machine learning (ML) strategies. An overview of ML strategies and their typical applications. ML comprises three main categories—supervised learning, unsupervised learning, and reinforcement learning.

wherein the model learns the mapping from the inputs to the outputs. The ultimate goal is to fit a model that understands the relationship between the inputs and the outputs. Examples of supervised learning strategies include regression modeling, artificial neural networks, decision trees, support vector machines, and random forests. On the other hand, unsupervised learning does not need as much human intervention as it does not require the dataset to be labeled. Unsupervised learning methods seek to explore patterns in datasets with no preexisting labels. Examples of unsupervised learning techniques include clustering, expectation maximization, generative adversarial networks, and deep belief networks. Reinforcement learning is an approach to choose the optimal action in a known/uncertain environment. It differs from supervised learning in that the labeled input/output pairs are not required and the suboptimal actions need not be explicitly corrected—those can simply be discarded. Here, the focus is on finding a balance between exploration (of uncharted territory) and exploitation (of current knowledge). Unlike standard virtual reality, which includes spatial environments, user navigation, and basic object interactions, XR environments can include virtual machinery, cameras and sensors, control software, human avatars and much more, creating a true digital twin of a corresponding physical system. Machine learning training data created synthetically in these systems replaces physical data that is difficult, costly or even impossible to acquire. In addition, use of synthetic data allows for *domain randomization* (Dr), a technique that has been shown to improve results over physical data alone, as it helps the neural network ignore spurious features in the training dataset (Scantlebury, Brown, Delft, & Deane, 2020).

15.4 Strategies to improve immersion and presence

15.4.1 Data augmentation

The need for training new abilities often requires that ML algorithms are presented with big data. The traditional approach was often to gather that data from users during the activity of interest and then train the ML models on this data. The need for massive data and relatively complex training scenarios for more and more realistic and feature-rich XR presents researchers with a problem that is often better solved by using *additional augmented data* for training these models, rather than relying solely on the real data. Data augmentation methods depend on the problem at hand. For visual tasks, these have included scaling of objects, translating, that is, moving objects spatially to various positions, rotation of objects at various angles, adding noise, altering the lighting conditions, changing the angle of view, etc. (Pai, 2017). For audio tasks, data augmentation often includes introducing deformations into a temporal dimension—by changing the audio speed, tonal frequencies, background noise, etc. Larger and more useful augmented datasets can be created using *adversarial learning* wherein a generative model competes against an adversary (Doya, 2000). The generative model with XR can also explore some states in the physical reality could be very costly and dangerous.

15.4.2 Reinforcement learning

Reinforcement learning is an area of ML concerned with how the digital twin of a user ought to take actions in an environment in order to maximize the notion of cumulative reward.

Generative adversarial networks (GANs) (Goodfellow, Pouget-Abadie, & Mirza, 2014) estimate such generative models via an adversarial process by training two models simultaneously: a generative model G to capture the data distribution, and a discriminative model D to estimate the probability that a sample comes from the training data but not the generative model G. Such an approach could be extended to XR by training a generative model G on virtual or simulated test data and then using a discriminative model D to estimate the probability that a sample comes from the real world (Espinosa Leal, Chapman, & Westerlund, 2019). In Vezhnevets et al. (Vezhnevets, Mnih, & Agapiou, 2016) a *strategic attentive writer* (STRAW), a deep recurrent neural network architecture, is proposed for learning high-level temporally abstracted macro-actions in an end-to-end manner based on observations from the environment. Macro-actions are sequences of actions commonly occurring. STRAW builds a multistep action plan, updated periodically based on observing rewards, and learns for how long to commit to the plan by following it without re-planning. Similar to GANs, STRAW could be used after the simulation learning stage so the user copes with any discrepancies between the simulation and the real world.

15.4.3 Algorithms and software implementations

An ML pipeline involves the following steps. We illustrate its use in the context of the stroke rehabilitation therapy using a smart wearable arm-band that has been described in detail in Yang, Deng, and Pang (2018) (Fig. 15.3) This stroke rehabilitation system is based on a *smart*

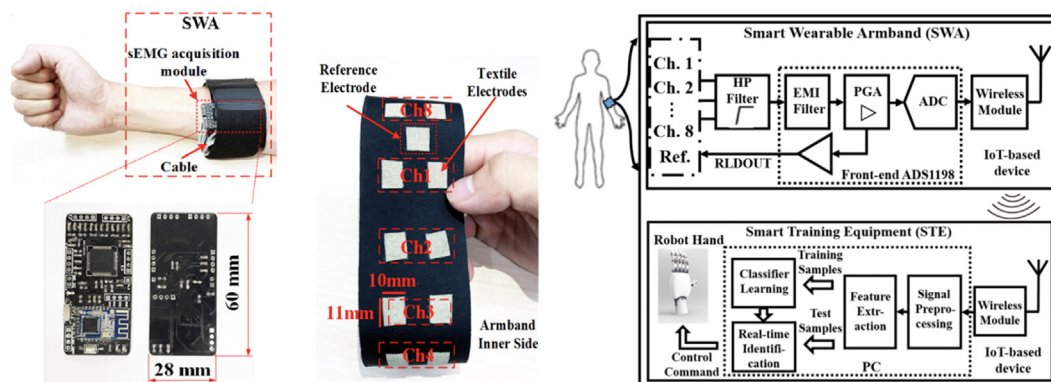


FIGURE 15.3 Machine learning-based stroke rehabilitation. An ML enhanced stroke rehabilitation therapy system demonstrated in [Yang, Deng, and Pang \(2018\)](#). This figure is created using two of the figures described there.

wearable armband (SWA), ML algorithms, and a 3-D printed dexterous robot hand. User comfort is one of the key issues which should be addressed for wearable devices. The SWA was developed by integrating a low-power and tiny-sized *Internet of Things* (IoT) sensing device with textile electrodes that can measure, preprocess, and wirelessly transmit bio-potential signals. By evenly distributing surface electrodes over the forearm of the user, some of the drawbacks of classification accuracy and poor performance can be mitigated. A new method was devised to find the optimal feature set. ML algorithms were leveraged to analyze and discriminate features of different hand movements, and their performances were appraised by classification complexity estimating algorithms and principal components analysis. According to the verification results, all nine gestures can be successfully identified with an average accuracy up to 96.20%. In addition, a 3-D printed five-finger robot hand was implemented for hand rehabilitation training purposes. The user's hand movement intentions were extracted and converted into a series of commands which were used to drive motors assembled inside the dexterous robot hand. The dexterous robot hand could mimic the user's gesture in a real-time manner, thereby proving that such a primitive XR system can be used as a training tool to facilitate rehabilitation process for the patients after stroke.

Some of the key steps in a generic ML pipeline are as follows:

1. *Data Collection and Data Loading*: The datasets are typically in comma-separated values and Excel formats. These are collected from the real-world and also using the data augmentation techniques mentioned earlier. In [Espinosa Leal et al. \(2019\)](#) the real world data was the time-series Electrocardiogram data, in particular the RR interval (RRI) data, that was collected through the SWA from 100 firefighters. It was broken down into 30 measurement windows prior to further processing or modeling.
2. *Exploratory Analysis*: An exploratory analysis is then performed using elementary statistical notions such as mean, standard deviation, and histograms to understand the granularity of the dataset, understand which of the data samples can be used reliably,

and to make a good choice of the target/predictor variables. The *heart rate variability* samples were explored to identify meaningful clusters. This was first attempted using classical clustering methods, such as the k-means clustering, but that attempt failed to identify any. So, complicated *convolutional autoencoders* (CAE) and *long short-term memory autoencoders* had to be used.

3. *Data Aggregation*: Some information might be scattered across different files. To overcome this limitation and to use additional threshold-based prefiltering, data aggregation techniques such as “grouping” are often used. In order to classify unseen data, a KNN classifier with the clusters and labels obtained from the validation results is used in [Yang et al. \(2018\)](#).
4. *Data Cleaning and Normalization*: The next step is to process the data to clean errors and inconsistencies using only “ex ante” information. This is a time-consuming process. There are two distinct techniques for data cleaning—Qualitative and Quantitative—to classify data errors. Qualitative techniques involve rules, constraints, and patterns to identify errors, for example, find and discard duplicate rows, find columns containing only one or no datapoint. Quantitative techniques employ statistical techniques to identify errors in the trained data. Once the errors are identified using these techniques, those are rectified by making changes in the script or by human intervention, or using a combination of both.
5. *Split Training, Validation, and Test*: Once we have a dataset consisting of the relevant features, it is split into Training, Validation, and Test sets. Usually, Training and Validation account for 70% of the data, while Test is the remaining 30%. This split is typically done randomly; however, there are several exceptions to this general rule. First, in classification problems, stratified sampling is often used if the minority class is significantly less frequent than the other class.
6. *Mutual Information Calculation*: Mutual information metric quantifies the mutual dependence between two variables. This metric is useful to describe nonlinear relationships between variables. Today, one can compute it using the “feature_selection.mutual_info_regression” function from *scikit-learn* between each feature.
7. *Identification of Informative Points*: To identify the most informative points, multiple iterations of the following procedure can be adopted: (1) Randomly sample n combinations from the dataset; (2) Train models on those points using the strategy of your choice – supervised learning or unsupervised learning or reinforcement learning; (3) Predict on the other points for scenario; (4) Obtain the average score on all scenarios; (5) Keep those combinations if this average is better than the current score.
8. *Performance Testing and Validation*: There are different methods for validating a predictive model—Leave-p-out cross validation, leave-one-out cross validation, and K-Fold cross validation. K-fold cross validation is a form of resampling method which means that samples are repeatedly drawn from a training set in order to refit the model and evaluate its performance or perhaps to select the level of flexibility of the model. This process is usually referred to as “model assessment.” The k-fold cross validation approach divides the training set into k groups that are equal. Initially, the primary fold is

the validation set or the hold-out set and a statistical learning method is applied to the remaining $k-1$ folds. For each model fitted to folds, responses for the validation set are being predicted. Afterwards, the *mean squared error* is calculated for the observations in the validation set. This is a repetitive process and occurs k times until every fold is treated as a validation set. K-Fold cross validation can be applied in various settings such as choosing the flexibility of a single statistical learning method or comparing different methods to each other. In Yang et al. (2018), a fivefold cross-validation scheme is used.

To implement the machine learning algorithms, commercial packages as well as freely available packages can be used. These include TensorFlow, Shogun, Apache Mahout, Apache Spark MLlib, Oryx 2, H2O.ai, Pytorch, RapidMiner, Weka, KNIME, Keras, and Scikit-learn (Martin, 2020; Top 11 Machine Learning Software – Learn before you regret!, 2020).

An example of a machine learning aided chatbot for XR is Woebot (Woebot, 2020). The Woebot platform uses machine learning and *natural language processing* (NLP) techniques to understand and evaluate a user's emotional state and cognitive ability so it can safely deliver an intervention at the right time. Its machine learning algorithm personalizes the experience according to the state of the user, the description of the problem at hand, and the case history of the user's progress through the program. Limited amounts of training data are available for many NLP tasks and that presents a challenge for data hungry deep learning methods. Given the high cost of annotating supervised training data, very large training sets are not available for most NLP tasks in XR. This problem can be addressed by implicitly performing limited transfer learning through the use of pretrained word embeddings such as those produced by word2vec (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013) or GloVe (Pennington, Socher, & Manning, 2014). Strong transfer task performance has also been demonstrated using pretrained sentence level embeddings (Conneau, Kiela, Schwenk, Barrault, & Bordes, 2017). In sentence encodings, the goal is to find a model that can map a sentence to a fixed-length vector representation. This vector encodes the meaning of the sentence and thus can be used for downstream tasks such as searching for similar documents as might be relevant in the context of the therapy. A way to achieve this goal is through a *universal sentence encoder* (USE) was proposed (Cer, Yang, & Kong, 2018). In USE, the idea is to design an encoder that summarizes any given sentence to a 512-dimensional sentence embedding. The same embedding is used to solve multiple tasks. Based on the mistakes it makes on those, the sentence embedding is updated. Since the same embedding has to work on multiple generic tasks, it will capture only the most informative features and discard noise. The intuition is that this will result in a generic embedding that transfers universally to a wide variety of NLP tasks such as relatedness, clustering, paraphrase detection, and text classification. Once the model is trained using the above tasks, it can map any sentence into fixed-length 512 dimension sentence embedding. This can be used for semantic search, paraphrase detection, clustering, smart-reply, text classification, and many other NLP tasks. Pretrained models for USE are available via Tensorflow Hub.

15.4.4 Conversational Agents

Chatbots enable a greater level of interactivity and engagement than and can be used without a therapist present. These use Natural Language Processing (NLP) to make sense of the

input so that the correct response can be given. When combined with speech recognition they can be used to simplify user interaction and deliver any form of therapy that can be defined through a set of rules or heuristics. The use of chatbots as general-purpose digital assistants is now established through products from Amazon, Google and Apple, where the conversational agent is combined with speech recognition.

As with XR, the first wave of medical applications have focused on mental health, delivering programs of support built around self-help Cognitive Behavioral Therapy (CBT) and related techniques. Woebot ([Woebot, 2020](#)) is a text-based mental health chatbot designed to treat depression and anxiety disorders in adults through CBT. The developers have shown that it can reduce symptoms of depression and anxiety when used over a period of 2 weeks ([Fitzpatrick, Darcy, & Vierhile, 2017](#)). Wysa ([Inkster, Sarda, & Subramanian, 2018](#); [Wysa - Everyday Mental Health, 2020](#)) is an AI-based “emotionally intelligent” bot, a virtual coach which uses CBT, dialectical behavior therapy, meditation, breathing, yoga, motivational interviewing and micro-actions to build mental resilience. In a randomised control trial (RCT) with 75 university students Tess, a text-messaging mental health chatbot ([Tess - Mental Health Chatbot, 2021](#)), was found to reduce depression by 13% and anxiety by 18% ([Fulmer, Joerin, Gentile, Lakerink, & Rauws, 2018](#)). Enjo is a chatbot designed to improve parental well-being through mindfulness. Although not marketed to treat mental health conditions an RCT with 28 participants showed it effectively reduced stress ([Ly, Ly, & Andersson, 2017](#)). Sleepio ([Sleepio and Can't sleep? Get to sleep, 2021](#)), a chatbot designed to improve sleep was also shown to reduce symptoms of anxiety and depression ([Health app: Sleepio for adults with poor sleep, 2017](#)). Youper is an emotional health assistant that claims to reduce stress after only 7 minutes of use ([Youper, 2021](#)).

More recently, Vigo ([Vigo, 2021](#)) is a digital assistant for stroke patients that uses a conversational chatbot and gamification to support both emotional and physical recovery. A small initial study has demonstrated feasibility and acceptance at the patient level ([Epalte, Tomsone, Vētra, & Bērziņa, 2020](#)). A further system developed at Edinburgh University, again tested on a small group of healthy users, showed that patient acceptance is high and that this technology can be used to support therapy ([Lohse, 2019](#)). The potential to integrate these technologies into Virtual Reality to provide a full package of rehabilitation therapy is clear.

15.4.5 Limitations

Research into the use of XR technologies in motor and cognitive rehabilitation has been conducted showing that the application of these technologies have the potential to improve and augment the standard rehabilitation process by increasing patient interaction, and promoting longer and more frequent therapy sessions. However, the therapy still requires assistance from another person, usually a therapy assistant or family member, to initiate and facilitate the therapy sessions

Until relatively recently, providing a fully immersive XR experience has been an expensive undertaking. The Computer Assisted Rehabilitation Environment, for example, combines a motion platform, an instrumented force plate dual belt treadmill, and a room scale VR

configuration to provide a high fidelity experience for rehabilitation, including for stroke survivors (De Luca, Manuli, & De Domenico, 2019). The initial and ongoing running costs are significant, however. Cost effective alternatives are now available and they support visual and audio very well. A limitation, particularly in VR solutions for physical rehabilitation, is to extend the experience to encompass force feedback. Haptic feedback is typically tactile at best (e.g., through vibrations in hand held devices). There are stylus/joystick devices available that will provide force feedback but these are rarely integrated into virtual environments for stroke rehabilitation. Haptic gloves are an expensive option and not currently widely available. AR solutions will be able to use real world objects for haptics and have more potential to develop effective force feedback as a part of the rehabilitation program. As for the other human senses of taste and smell, these have not yet been integrated into XR outside of research laboratories.

Use of Machine Learning and other artificial intelligence techniques are becoming more widespread in clinical applications. Radiology is leading the way with such approaches being applied to automate diagnosis from medical images. It is particularly effective when integrated into the clinical workflow as a tool to assist radiologists. Other specialties too are benefiting, particularly as an addition to any treatment that is fairly procedural and transparent. The full impact has not yet been realized, however, and deep learning techniques will be able to provide new and innovative software approaches for a broader spectrum of treatments.

It is possible that the implementation of intelligent software within XR therapy applications could be used to further improve their feasibility and efficacy, for example, by reducing the need for external assistance once the session is started or automatically adjusting the session's difficulty based on the performance of the user.

15.4.6 Ethical issues

XR technology has the potential to impact rehabilitation at a fundamental level. Despite all the promised benefits, however, it has also created a host of potential adverse effects and interesting and important ethical conundrums of which the users need to be aware. These risks can broadly be summarized as follows: (1) motion sickness; (2) risk of data safety (3) information overload; (4) risks arising from the phenomena of superrealism (Behr, Nosper, Klimmt, & Hartmann, 2005; Slater, Gonzalez-Liencre, & Haggard, 2020; Spiegel, 2018). There is a limited scope of discussion here and we will discuss only (1) and (2), which are pertinent to rehabilitation.

Motion sickness is quite commonly reported in VR (it tends not to be an issue for AR), in some reports up to 70% of users, and is caused by a mismatch between the expected feedback from the users' perceptual system and the efferent feedback provided by the VR system. There are other factors which can also increase risk of motion sickness such as visual display characteristics (frame rate/field-of-view), vection, context and intuitiveness of interaction, presence of navigation control and past gaming experience. The effects are minimized if the user is stationary in the real world and is only carrying out tasks that can be reached from

that position, that is, no need to walk around. However, there is a need for more research in elderly population which is relevant in rehabilitation.

The use of specialised XR equipment in a hospital will also be an issue for ethical consideration. The equipment, for example, a VR headset, will have to conform to the infection prevention and control criteria that will be in place. Sharing of equipment between patients will therefore have to follow strict guidelines. In the case of a headset, normal methods of disinfecting will not be possible due to potentially damaging internal electronics. There are alternatives (e.g., a system is available for VR headsets that uses Ultraviolet C directional light to destroy all bacteria, fungi, and viruses) but it may be necessary to budget individual equipment for each patient.

The other major ethical controversy arises from the phenomenon variably described as *superreality*. It arises from intensification of experience as any feeling may be intensified in a XR environment, and thereby potentially straining the participants' coping abilities and creating adverse cognitive, emotional and behavioral disturbances. These may further be exacerbated after re-entry into the real world following the XR experience. Thus, XR poses potential mental health risks, including Depersonalization/Derealization Disorder and raises serious concerns related to personal neglect of users' own actual bodies and real physical environments. This might be stressful and detrimental to someone with poststroke disability. Finally, there are other moral and social risks associated with the way VR blurs the distinction between the real and illusory. The answers are not clear yet regarding these concerns about XR and further research is required.

15.5 Conclusions and future direction

Conventional stroke rehabilitation is a complex process, involving multiple disciplines of healthcare professionals seamlessly transferring care from one another or working together as a team. In order to become a mainstream rehabilitation process, AI-based rehabilitation techniques should adopt the same principles of integrated approach rather than targeting one or the other specific areas of the impairment. Despite having a huge potential for delivering a bigger dose of therapy in every aspect of stroke rehabilitation and thereby improving morbidity, quality of life, and possibly reducing the overall cost of care, XR, and/or robotics based rehabilitation techniques are still in its infancy and its integration with artificial intelligence is virtually nonexistent. A lack of integration between the key healthcare professionals with the experts in these fields along with a general inertia in rehabilitation research and lack of interest from the big tech companies are a few main reasons behind this lack of progress in this area. However, if proven to be successful, a XR-based affordable "total rehabilitation unit," administered by the members of the family or a semi trained therapy assistant, supported by the conventional trained therapists, with a fully integrated artificial intelligence system continuously learning to decide the right program and platform individualized to the patients' needs and delivering every aspect of rehabilitation with a tailor-made dose tolerated by the individual, could be a game-changer for millions of stroke victims all around the

globe. This system would be particularly important for the developing countries where the fully trained neurotherapists are not easily accessible for the majority of these patients.

References

- Behr, K. M., Nosper, A., Klimmt, C., & Hartmann, T. (2005). Some practical considerations of ethical issues in VR research. *Presence: Teleoperators and Virtual Environments*, 14(6), 668–676. Available from <https://doi.org/10.1162/105474605775196535>.
- Bernhardt, J., Dewey, H., Thrift, A., & Donnan, G. (2004). Inactive and Alone: Physical Activity within the First 14 Days of Acute Stroke Unit Care. *Stroke; a Journal of Cerebral Circulation*, 35(4), 1005–1009. Available from <https://doi.org/10.1161/01.STR.0000120727.40792.40>.
- Boone, A. E., Wolf, T. J., & Engsberg, J. R. (2019). Combining virtual reality motor rehabilitation with cognitive strategy use in chronic stroke. *American Journal of Occupational Therapy*, 73(4). Available from <https://doi.org/10.5014/ajot.2019.030130>.
- Cameirão, Pereira, M. S. Badia, F. S. B. i. (2017). Virtual reality with customized positive stimuli in a cognitive-motor rehabilitation task. *2017 International Conference on Virtual Rehabilitation (ICVR)*. 1–7. Available from <https://doi.org/10.1109/ICVR.2017.8007543>.
- Conneau, A., Kiela, D., Schwenk, H., Barrault, L., Bordes, A. (2017). Supervised learning of universal sentence representations from natural language inference data. *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*.
- Cer, D., Yang, Y., Kong, S., et al. (2018). Universal sentence encoder. *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*.
- Dascal, J., Reid, M., Ishak, W. W., et al. (2017). Virtual reality and medical inpatients: A systematic review of randomized, controlled trials. *Innovations in Clinical Neuroscience*, 14(1–2), 14–21. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5373791/pdf/icns_14_1-2_14.pdf.
- da Silva, C. A., Fernandes, A. R., & Grohmann, A. P. (2015). Star: Speech therapy with augmented reality for children with autism spectrum disorders. *Lecture Notes in Business Information Processing*, 227, 379–396. Available from https://doi.org/10.1007/978-3-319-22348-3_21.
- de Assis, G. A., Corrêa, A. G. D., Martins, M. B. R., Pedrozo, W. G., & Lopes, Rd. D. (2016). An augmented reality system for upper-limb post-stroke motor rehabilitation: a feasibility study. *Disability and Rehabilitation: Assistive Technology*, 11(6), 521–528. Available from <https://doi.org/10.3109/17483107.2014.979330>.
- De Luca, R., Manuli, A., De Domenico, C., et al. (2019). Improving neuropsychiatric symptoms following stroke using virtual reality: A case report. *Medicine*, 98(19), e15236. Available from <https://doi.org/10.1097/MD.00000000000015236>.
- Doya, K. (2000). Reinforcement learning in continuous time and space. *Neural Computation*, 12(1), 219–245. Available from <https://doi.org/10.1162/089976600300015961>.
- Duncan, P. W., Goldstein, L. B., Horner, R. D., Landsman, P. B., Samsa, G. P., & Matchar, D. B. (1994). Similar motor recovery of upper and lower extremities after stroke. *Stroke; a Journal of Cerebral Circulation*, 25(6), 1181–1188. Available from <https://doi.org/10.1161/01.STR.25.6.1181>.
- Epalte, K., Tomsone, S., Vētra, A., & Bērziņa, G. (2020). Patient experience using digital therapy “Vigo” for stroke patient recovery: A qualitative descriptive study. *Disability and Rehabilitation: Assistive Technology*, 1–10. Available from <https://doi.org/10.1080/17483107.2020.1839794>.
- Escalona, F., Martinez-Martin, E., Cruz, E., Cazorla, M., & Gomez-Donoso, F. (2020). EVA: EVALuating at-home rehabilitation exercises using augmented reality and low-cost sensors. *Virtual Reality*, 24, 567–581. Available from <https://doi.org/10.1007/s10055-019-00419-4>.

- Espinosa Leal, L. A., Chapman, A., Westerlund, M. (2019). Reinforcement learning for extended reality: designing self-play scenarios. <https://scholarspace.manoa.hawaii.edu/bitstream/10125/59456/0016.pdf>. Accessed June 19, 2021.
- Faria, A. L., Andrade, A., Soares, L., & i Badia, S. B. (2016). Benefits of virtual reality based cognitive rehabilitation through simulated activities of daily living: a randomized controlled trial with stroke patients. *Journal of Neuroengineering and Rehabilitation*. Available from <https://doi.org/10.1186/s12984-016-0204-z>.
- Faria, A. L., Cameirão, M. S., Couras, J. F., Aguiar, J. R. O., Costa, G. M., & Bermúdez i Badia, S. (2018). Combined cognitive-motor rehabilitation in virtual reality improves motor outcomes in chronic stroke - A pilot study. *Frontiers in Psychology*, 9(MAY). Available from <https://doi.org/10.3389/fpsyg.2018.00854>.
- Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. *JMIR Mental Health*, e19. Available from <https://doi.org/10.2196/mental.7785>.
- Fulmer, R., Joerin, A., Gentile, B., Lakerink, L., & Rauws, M. (2018). Using psychological artificial intelligence (TESS) to relieve symptoms of depression and anxiety: Randomized controlled trial. *JMIR Mental Health*, e64. Available from <https://doi.org/10.2196/mental.9782>.
- Giachero, A., Calati, M., Pia, L., et al. (2020). Conversational therapy through semi-immersive virtual reality environments for language recovery and psychological well-being in post stroke aphasia. *Behavioural Neurology*, 2020. Available from <https://doi.org/10.1155/2020/2846046>.
- Gittins, M., Vail, A., Bowen, A., et al. (2020). Factors influencing the amount of therapy received during inpatient stroke care: an analysis of data from the UK Sentinel Stroke National Audit Programme. *Clinical Rehabilitation*, 34(7), 981–991. Available from <https://doi.org/10.1177/0269215520927454>.
- Global, regional, and national burden of stroke, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. (2019). *The Lancet Neurology*. 18(5) 439–458. Available from [https://doi.org/10.1016/S1474-4422\(19\)30034-1](https://doi.org/10.1016/S1474-4422(19)30034-1).
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2672–2680.
- Health app: Sleepio for adults with poor sleep. <https://www.nice.org.uk/advice/mib129/chapter/Summary>. Published 2017. Accessed June 19, 2021.
- Inkster, B., Sarda, S., & Subramanian, V. (2018). An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: Real-world data evaluation mixed-methods study. *JMIR mHealth and uHealth*, 6(11). Available from <https://doi.org/10.2196/12106>.
- John N. W., Day T. W., Pop S. R., et al. (2019). Virtual reality environment for the cognitive rehabilitation of stroke patients. 2019 11th International Conference on Virtual Worlds and Games for Serious Applications, VS-Games 2019 - Proceedings. 1. Available from <https://doi.org/10.1109/VS-Games.2019.8864513>.
- Kim, W.-S., Cho, S., Ku, J., et al. (2020). Clinical application of virtual reality for upper limb motor rehabilitation in stroke: Review of technologies and clinical evidence. *Journal of Clinical Medicine*, 3369. Available from <https://doi.org/10.3390/jcm9103369>.
- Krakauer, J., & Carmichael, S. T. (2017). Chronic hemiparesis: Motor learning, compensation, and the challenge of reversing impairment in late stroke. *Broken Movement: The Neurobiology of Motor Recovery after Stroke*.
- Langhorne, P., & Ramachandra, S. (2020). Organised inpatient (stroke unit) care for stroke: network meta-analysis. *Cochrane Database of Systematic Reviews*, 2020(4). Available from <https://doi.org/10.1002/14651858.CD000197.pub4>.
- Langhorne, P., Wu, O., Rodgers, H., Ashburn, A., & Bernhardt, J. (2017). A very early rehabilitation trial after stroke (AVERT): a phase III, multicentre, randomised controlled trial. *Health Technology Assessment*, 21(54), 1–119. Available from <https://doi.org/10.3310/hta21540>.
- Laver, K. E., Lange, B., George, S., Deutsch, J. E., Saposnik, G., & Crotty, M. (2017). Virtual reality for stroke rehabilitation. *Cochrane Database of Systematic Reviews*, 2017(11). Available from <https://doi.org/10.1002/14651858.CD008349.pub4>.

- Legg, L. A., Drummond, A. E., & Langhorne, P. (2006). Occupational therapy for patients with problems in activities of daily living after stroke. *Cochrane Database of Systematic Reviews* (4). Available from <https://doi.org/10.1002/14651858.CD003585.pub2>.
- Lohse, K.-M. (2019). *Designing a Prototype Chatbot for People with Minor Strokes*.
- Ly, K. H., Ly, A. M., & Andersson, G. (2017). A fully automated conversational agent for promoting mental well-being: A pilot RCT using mixed methods. *Internet Interventions*, 10, 39–46. Available from <https://doi.org/10.1016/j.invent.2017.10.002>.
- Maggio, M. G., Latella, D., Maresca, G., et al. (2019). Virtual reality and cognitive rehabilitation in people with stroke: An overview. *Journal of Neuroscience Nursing*, 51(2), 101–105. Available from <https://doi.org/10.1097/JNN.0000000000000423>.
- Maggio, M. G., Maresca, G., De Luca, R., et al. (2019). The growing use of virtual reality in cognitive rehabilitation: Fact, fake or vision? a scoping review. *Journal of the National Medical Association*, 111(4), 457–463. Available from <https://doi.org/10.1016/j.jnma.2019.01.003>.
- Marshall, J., Devane, N., Talbot, R., et al. (2020). A randomised trial of social support group intervention for people with aphasia: A novel application of virtual reality. *PLoS One*, 15(9), e0239715. Available from <https://doi.org/10.1371/journal.pone.0239715>.
- Martin, S. (2020). 10 Most Popular Machine Learning Software Tools in 2020 (updated). <https://towardsdatascience.com/10-most-popular-machine-learning-software-tools-in-2019-678b80643ceb>. Accessed June 19, 2021.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G., Dean, J. (2013). Distributed representations of words and phrases and their compositionality. Proceedings of NIPS'13.
- Pai, P. (2017). Data Augmentation Techniques in CNN using Tensorflow. <https://medium.com/ymedialabs-innovation/data-augmentation-techniques-in-cnn-using-tensorflow-371ae43d5be9>. Accessed June 19, 2021.
- Pennington J., Socher R., Manning C.D. Glove: Global vectors for word representation. Proceedings of EMNLP. 2014.
- Riva, G., Mancuso, V., Cavedoni, S., & Stramba-Badiale, C. (2020). Virtual reality in neurorehabilitation: a review of its effects on multiple cognitive domains. *Expert Review of Medical Devices*, 17(10), 1035–1061. Available from <https://doi.org/10.1080/17434440.2020.1825939>.
- Rohrbach, N., Gulde, P., Armstrong, A. R., et al. (2019). An augmented reality approach for ADL support in Alzheimer's disease: A crossover trial. *Journal of Neuroengineering and Rehabilitation*, 16(1). Available from <https://doi.org/10.1186/s12984-019-0530-z>.
- Royal College of Physicians London. (2016). Stroke Guidelines. <https://www.rcplondon.ac.uk/guidelines-policy/stroke-guidelines>. Accessed June 19, 2021.
- Scantlebury, J., Brown, N., Delft, F. V., & Deane, C. M. (2020). Dataset augmentation allows deep learning-based virtual screening to better generalize to unseen target classes, and highlight important binding interactions. *bioRxiv*. Available from <https://doi.org/10.1101/2020.03.06.979625>.
- Schneider, E. J., Lannin, N. A., Ada, L., & Schmidt, J. (2016). Increasing the amount of usual rehabilitation improves activity after stroke: A systematic review. *Journal of Physiotherapy*, 62(4), 182–187. Available from <https://doi.org/10.1016/j.jphys.2016.08.006>.
- Shakespeare, T., & Officer, A. (2011). World report on disability. *Disability and Rehabilitation*, 33.
- Slater, M., Gonzalez-Liencre, C., Haggard, P., et al. (2020). The ethics of realism in virtual and augmented reality. *Frontiers in Virtual Reality*, 1. Available from <https://doi.org/10.3389/frvir.2020.00001>.
- Sleepio | Can't sleep? Get to sleep. <https://www.sleepio.com/>. Accessed June 19, 2021.
- Song, X., Ding, L., Zhao, J., Jia, J., Shull, P. (2019). Cellphone augmented reality game-based rehabilitation for improving motor function and mental state after stroke. *2019 IEEE 16th International Conference on Wearable and Implantable Body Sensor Networks, BSN 2019 - Proceedings*. Available from <https://doi.org/10.1109/BSN.2019.8771093>.

- Spiegel, J. S. (2018). The ethics of virtual reality technology: Social hazards and public policy recommendations. *Science and Engineering Ethics*, 24(5), 1537–1550. Available from <https://doi.org/10.1007/s11948-017-9979-y>.
- Tess - Mental Health Chatbot. <https://www.x2ai.com/>. Accessed June 19, 2021.
- Tieri, G., Morone, G., Paolucci, S., & Iosa, M. (2018). Virtual reality in cognitive and motor rehabilitation: facts, fiction and fallacies. *Expert Review of Medical Devices*, 15(2), 107–117. Available from <https://doi.org/10.1080/17434440.2018.1425613>.
- Top 11 Machine Learning Software – Learn before you regret! (2020). <https://data-flair.training/blogs/machine-learning-software/>. Accessed June 19, 2021.
- Vezhnevets, A., Mnih, V., Agapiou, J., et al. (2016). Strategic attentive writer for learning macro-actions. *Advances in Neural Information Processing Systems*, 3494–3502.
- Vigo. <https://vigo.health/en/>. Accessed June 19, 2021.
- WHO. (2020). Rehabilitation. <https://www.who.int/news-room/fact-sheets/detail/rehabilitation>. Accessed June 19, 2021.
- Woebot. (2020). <https://woebothealth.com/>. Accessed June 19, 2021.
- Wysa - Everyday Mental Health. (2020). <https://www.wysa.io/>. Accessed June 19, 2021.
- Yang, G., Deng, J., Pang, G., et al. (2018). An IoT-Enabled Stroke Rehabilitation System Based on Smart Wearable Armband and Machine Learning. *IEEE Journal of Translational Engineering in Health and Medicine*, 6. Available from <https://doi.org/10.1109/JTEHM.2018.2822681>.
- Youper. <https://www.youper.ai/>. Accessed June 19, 2021.